

POWERSHIELD TECHNOLOGIES PVT. LTD.

Technical Submittal — PS-1500-VFI UPS System | Project: VERTEX Hyperscale Data Centre, Pune | Submittal No: PS-VERTEX-SUB-001 Rev 1 | Date: 28 March 2025

1. Product Overview

The PowerShield PS-1500-VFI is a three-phase, double-conversion online UPS designed for mission-critical applications. Manufactured at our ISO 9001:2015 certified facility in Navi Mumbai. This submittal is provided in response to Project VERTEX specification VERTEX-ELEC-SPEC-UPS-001 Rev 2.

2. Technical Specifications — PS-1500-VFI

Parameter	Specification
Rated Output Capacity	1500 kVA / 1350 kW (0.9 pf)
Input Voltage	Three-phase, 415V AC $\pm 10\%$, 50 Hz
Input Frequency	50 Hz $\pm 5\%$
Input THDi at Full Load	2.8% (with active front-end rectifier)
Output Voltage	Three-phase, 415V AC $\pm 1\%$
Output Frequency	50 Hz $\pm 0.1\%$
Output THDu at Full Load	0.8%
Efficiency — Online Mode, 100% Load	95.8%
Efficiency — ECO Mode	98.2%
Transfer Time to Static Bypass	8 ms
Battery Technology	VRLA AGM, sealed cells
Battery Autonomy at Full Load	8 minutes (standard configuration)
Battery Design Life	10 years at 20°C
Enclosure Ingress Protection	IP20
IEC 62040-3 Classification	VFI-SS-111
Certifications	CE, BIS IS 16315, IEC 62040-3

3. Efficiency Note

The PS-1500-VFI achieves 95.8% efficiency in double-conversion online mode at 100% rated load. ECO mode (line-interactive) achieves 98.2% but is recommended for non-critical applications only and does not apply to VFI-mode Tier IV operation.

4. Battery System Details

The standard PS-1500-VFI configuration provides 8 minutes of autonomy at full rated load (1500 kVA, 0.9 pf). An extended battery cabinet (PS-EBC-15) is available as an optional accessory to achieve 10-minute or greater autonomy. Additional lead time for PS-EBC-15: 6 weeks from order confirmation.

5. Enclosure Protection Note

Standard PS-1500-VFI enclosures are rated IP20, suitable for clean, dry indoor environments such as purpose-built electrical rooms with controlled access. An IP31-rated enclosure variant is available as a built-to-order option with 4 additional weeks lead time.